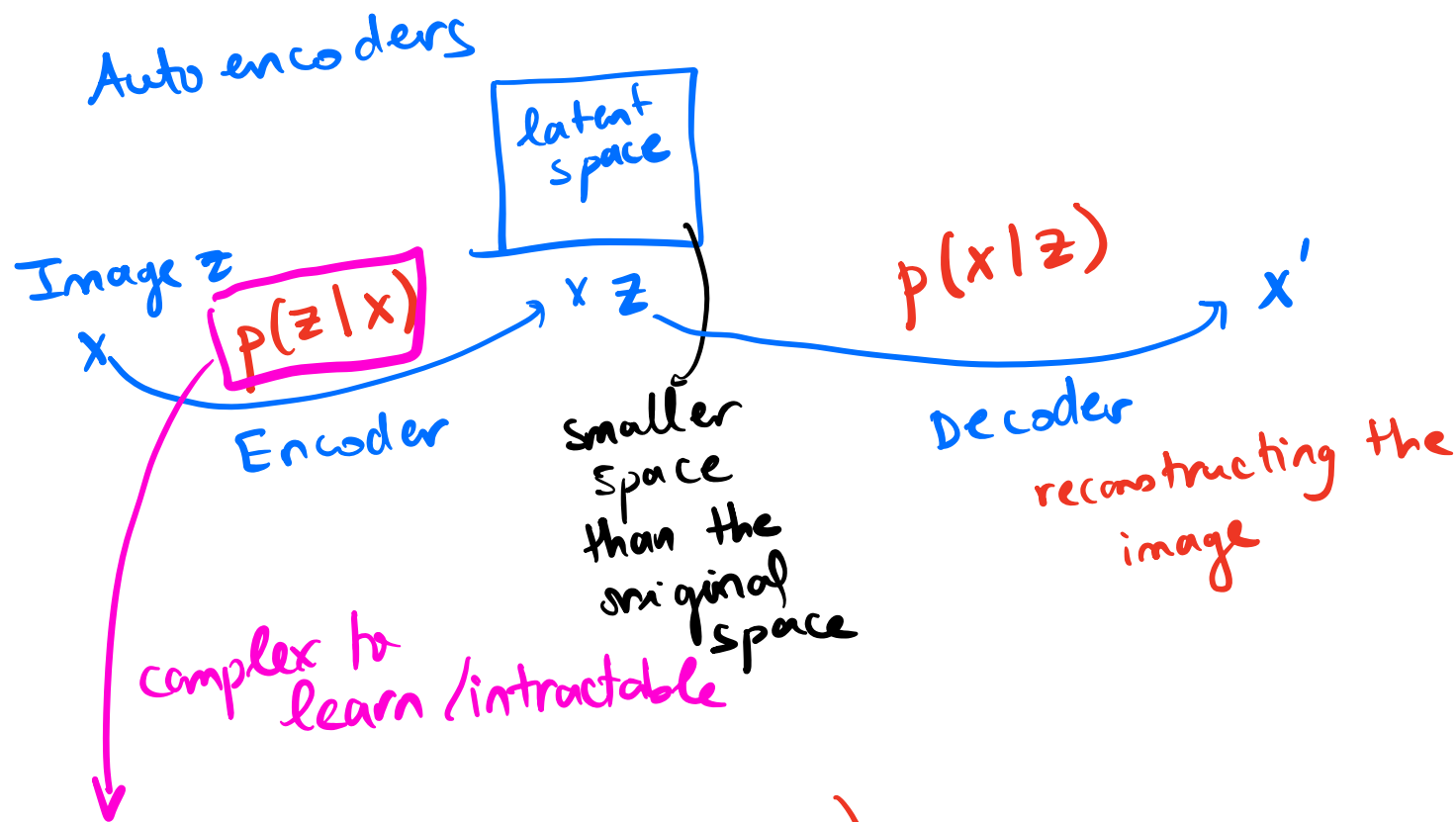


* Review from last time

Variational autoencoders



$$p(z|x) = \frac{p(x|z) \cdot p(z)}{\boxed{p(x)}}$$

" $\int p(x, z) dz$) infinite sum intractable in terms of computation

variational inference:

$$q_{\phi}(z|x) \sim \underbrace{N(\mu_{\phi}(x), \sigma_{\phi}(x))}$$

why this choice?

loss function:

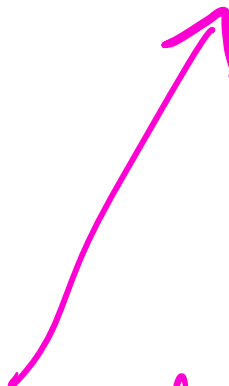
* reconstruction loss

$$\mathbb{E}_{z \sim q_{\phi}(z|x)} [\log p(x|z)]$$

* KL divergence term

$$KL(\underbrace{q_{\phi}(z|x)}, p(z))$$


$$p(\mathbf{z}) = N(\mathbf{0}, \mathbf{I})$$



choice of gaussian (normal) * simple to sample from

$KL(\text{gaussian}, \text{gaussian})$ * making sure that we have some spread in the latent values to learn correctly different features

which can be computed efficiently.

total loss =

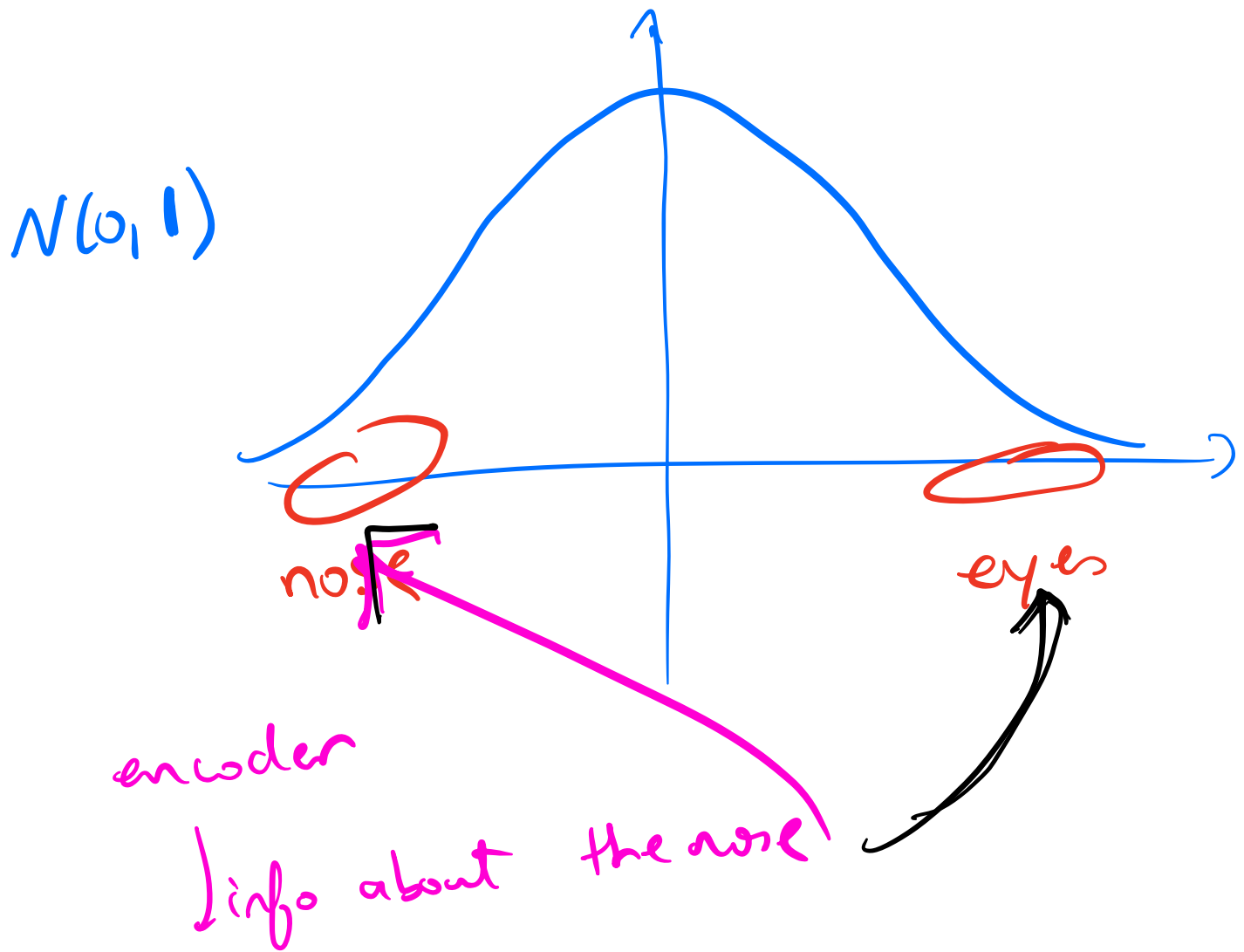
$$\mathbb{E}_{z \sim q_{\phi}(z|x)} [\log p(x|z)]$$

$$= \text{KL}(q_{\phi}(z|x), p(z))$$

this is a lower bound
to the log likelihood
function given that
we're using $q_{\phi}(z|x)$

an approximation of
 $p(z|x)$

not exact one.



choice of encoder :

$$N(\mu_{\Phi}(x), \sigma_{\Phi}^2(x))$$



$$z \sim N(\mu_\phi(x), \sigma_\phi^2(x))$$

troublesome for the learning of the model

$$\epsilon \sim N(0, 1)$$

$$N(\mu^{(x)}, \sigma^{(x)})$$

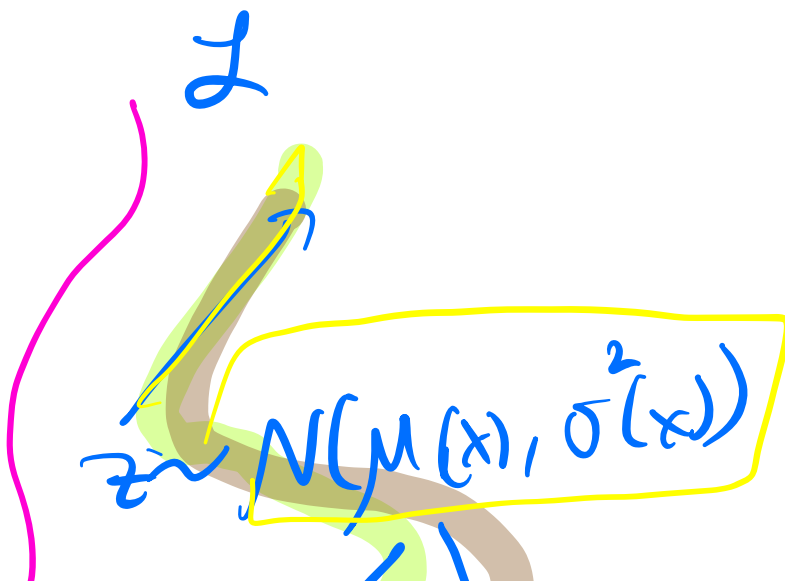
$$z = \sigma_\phi(x) \cdot \epsilon + \mu_\phi(x)$$

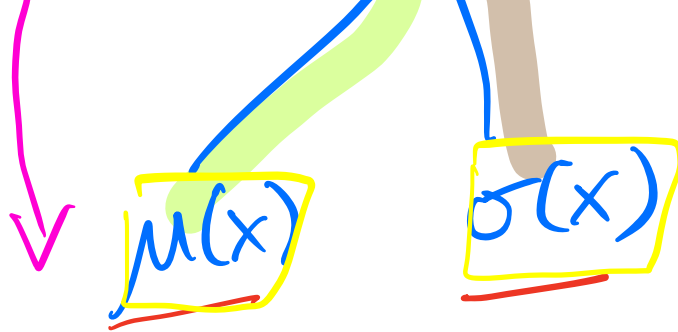
reparametrization trick

$$E(z) = \mu_\phi(x)$$

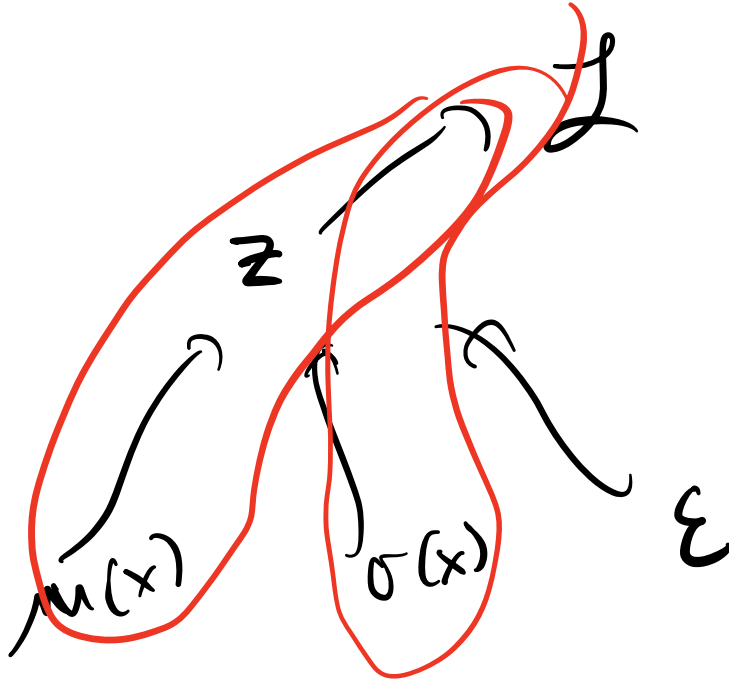
$$\text{Var}(z) = \text{Var}(\sigma_\phi(x) \cdot \epsilon)$$

$$= (\sigma_\phi(x))^2 \cdot \text{Var}(\epsilon) = \sigma_\phi^2(x)$$





$$\boxed{\frac{\partial z}{\partial \mu}}$$



deterministic
path from
 $\mu(x)$ to L
so won't have
the "random"
derivatives
anymore

reparam trick applies to
normal distributions.

(not to all distributions though)

MNIST Dataset

Generate images of digits using
a VAE

Canvas / Files) Problems in Class

download folder VAE

1) sample-gaussian function in utils.py
Implement the reparam trick
`torch.randn()`

2) Implement the
negative-elfo-bound in vae.py

$$\mathbb{E}_{z \sim q_{\phi}(z|x)} [\log p(x|z)] \quad \text{intractable}$$

$$\approx -\log p(x|z^{(i)})$$

where $z^{(i)}$ is one sample from $q_{\phi}(z|x)$